

A protein- and phosphoprotein-level test of the spaceflight kidney matrix-high / DCT-low remodeling signal

OSD-462 / RR-10 multi-omics anchor for the RRRM-2 transcriptomic axis

RRRM-2 Kidney Transcriptome Project

Compendium generated 2026-05-21
run run_20260521_osd462_anchor

Abstract

The RRRM-2 (OSD-771) kidney transcriptome analysis identified a recurrent ISS-transit remodeling axis: an extracellular-matrix (ECM) program elevated in flight and a distal-convoluted-tubule (DCT) Na-Cl transport program (WNK-SPAK/OSR1-NCC) reduced in flight. Because an RNA co-expression shift is weak standalone evidence, we used the independent OSD-462 / RR-10 spaceflight kidney multi-omic study (TMT proteomics and phosphoproteomics, female B6129SF2/J, single ISS-transit-type arm) as a protein/phosphoprotein *anchor*. We pre-registered four hypotheses and explicit falsification rules and tested them with abundance/peptide-matched random-gene-set nulls. **Findings.** (i) The matrix-high / DCT-low direction *recurs at the RNA level* in OSD-462 (pathway-vector cosine = 0.87, 95% CI [0.65, 0.90]; gate **PASS**). (ii) It does *not* translate into protein-abundance concordance: no targeted set is concordant in the predicted direction, and matrix proteins move significantly *opposite* to their transcripts (two-sided matched-null $p = 0.012$). (iii) However, the NCC activating phosphorylation is specifically and strongly suppressed — the N-terminal regulatory threonine cluster (mean = $-0.83 \log_2$; 7 sites $p < 0.05$) and the upstream SPAK activating sites (mean = -0.41) rank among the most down-regulated phosphosites in the entire kidney phosphoproteome — while total NCC protein is unchanged ($+0.09$). (iv) Network-nominated candidate genes are *not* enriched among protein- or phospho-changing genes. **Interpretation.** The transcriptomic remodeling signal is real and reproducible, but its protein-level correlate is a change in *transport-pathway signaling activity* (reduced NCC phosphoactivation), not in transporter or matrix *abundance*. The claim is upgraded from “a co-expression network shifted” to “a transcript- and signaling-concordant suppression of the WNK-SPAK-NCC activation arm,” while the matrix arm is shown to be transcript-level only.

1 Purpose and design

This compendium reports an independent multi-omics test of one transcriptomic result. The RRRM-2 backbone analysis (run run_20260519_000547_2500g) is a *fixed input*, not re-derived here. OSD-462 / RR-10 is the NASA kidney multi-omic study: female B6129SF2/J mice, left kidney, single terminal collection arm, with RNA-seq, TMT proteomics, and phosphoproteomics. Its hard constraints shape every claim (Table 1): it corroborates only the *ISS-transit* side (no late-adapting-return arm, no aged animals); the comparison is cross-study, cross-strain and cross-modality, so the unit of inference is the *direction* of the flight effect (Space Flight — Ground Control), never sample-level pooling; and bulk whole-kidney mass spectrometry dilutes the small DCT compartment. Primary inference is an abundance/peptide-matched random-gene-set null, which is robust to the generically weak RNA $\leftrightarrow$ protein correlation and to gene-set

size.

Property	OSD-462 / RR-10 (anchor)	RRRM-2 / OSD-771 (reference)
Tissue	Left kidney	Kidney
Sex	Female	Female
Strain	B6129SF2/J (WT)	C57BL/6NTac
Age	14–15 wk (single)	Young + Old
Arms	Single terminal arm	ISS-T + LAR
Modalities	RNA-seq, TMT proteomics, phosphoproteomics	RNA-seq only

Table 1: Cohort comparison. OSD-462 can corroborate the ISS-transit flight signal only; the late-adapting-return and age-stratified findings remain RNA-only and out of scope here.

Estimation. Protein flight effects are $\text{mean}(\log_2 \text{FL}) - \text{mean}(\log_2 \text{GC})$ computed *within* each TMT plex (“Samp1-5”, “Samp6-10”) on row-normalized S/N, with per-channel median (sample-loading) centering, then averaged across plexes; this removes the per-plex batch constant. Genes are required to be quantified in both plexes with ≥ 2 peptides for the primary tests (≥ 4 -peptide sensitivity reproduces the conclusions). Phosphosite effects use a per-site linear model $\log_2(\text{S/N}) \sim \text{flight} + \text{plex}$, reporting the flight coefficient with a t -based 95% CI. The RNA recurrence gate compares pathway-effect vectors (mean per-gene flight effect per mechanism gene set) by cosine, with a sample-resampling bootstrap CI and leave-one-pathway-out check, mirroring the OSD-513 cross-study test.

2 Pre-registered hypotheses and outcomes

H1 – matrix protein concordance

not supported. Matrix/ECM proteins do not move in the RNA-predicted (up) direction; they move significantly down (Table 3).

H2 – DCT protein concordance

not supported. DCT/WNK/NCC *abundance* is flat (NCC +0.09 $\log_2$); not concordant with the predicted decrease.

H3 – phospho activity

PASS. The WNK–SPAK/OSR1–NCC activating phosphorylation is specifically and strongly reduced in flight (Table 5).

H4 – network translation

not supported. LIONESS/node2vec candidates are not enriched among protein- or phospho-changing genes (Table 6).

3 Results

3.1 Layer 4 — the RNA signal recurs (gate **PASS**)

Before any cross-modality claim, we confirmed that OSD-462’s own RNA flight effect reproduces the RRRM-2 ISS-transit direction. The pathway-effect vectors align with cosine 0.869 (95% CI [0.648, 0.903]), robust to dropping any single pathway (leave-one-out range [0.823, 0.904]), with 10/11 pathways sign-concordant (Table 2, Fig. 2A). The ECM program is up in both cohorts (OSD-462 +0.22, RRRM-2 +0.28) and the DCT/NCC/WNK program is down in both (−0.13 vs −0.08). The gate passes, so a protein-level result is interpretable.

Pathway	OSD-462 RNA	RRRM-2 RNA	sign agree
preservation_stress_response	0.157	0.293	yes
ecm_organization	0.222	0.280	yes
mmp_adam_proteolysis	0.197	0.123	yes
slp_slpr3	0.123	0.106	yes
tlr4_innate	0.075	0.095	yes
macrophage_inflammation	0.109	0.090	yes
integrin_cell_adhesion	0.119	0.087	yes
tubular_transport_broad	-0.048	0.042	no
fibrosis_tgfb_emt	0.149	0.025	yes
oxidative_stress_nrf2	0.021	0.005	yes
dct_ncc_wnk_transport	-0.130	-0.075	yes

Table 2: Layer 4 RNA recurrence: pathway-level flight effects (mean per-gene SF–GC / FLT–GC $\log_2$). The matrix-high / DCT-low direction recurs; cosine = 0.869, 95% CI [0.648, 0.903], 11 pathways, 10/11 sign-concordant.

3.2 Layer 1 — the signal does not reach protein abundance

The genome-wide RRRM-2-RNA $\leftrightarrow$ OSD-462-protein flight-effect correlation is weak (Spearman 0.043, Pearson 0.030; $n = 6,418$ genes), as expected for cross-study bulk TMT. Against the matched null, *no* targeted set is concordant in its predicted direction (all Benjamini–Hochberg $q \geq 0.92$; Table 3, Fig. 1A). The matrix set is in fact *anti*-concordant: mean protein effect $-0.107 \log_2$ where “up” was predicted (two-sided matched-null $p = 0.012$), with 14/15 matrix genes showing transcript-up but protein-down and a within-set RNA $\leftrightarrow$ protein Spearman of -0.396 . The DCT set is flat (-0.044), driven by NCC protein being maintained or slightly elevated (Fig. 1B). Representative machinery and peptide depth are in Table 4.

Gene set	n	pred.	mean prot.	$p_{\uparrow}$	concord.	p_c	sign agr.
ecm_organization	15	up	-0.107	0.993	-0.396	0.917	0.13
dct_ncc_wnk_transport	10	down	0.044	0.959	0.018	0.531	0.60
tlr4_innate	8	up	-0.021	0.797	-0.405	0.881	0.25
slp_slpr3	9	up	-0.045	0.946	0.167	0.350	0.56
tubular_transport_broad (control)	15	up	0.043	0.069	0.071	0.398	0.67

Table 3: Layer 1 protein-abundance concordance. “mean prot.” is the mean OSD-462 protein flight effect; $p_{\uparrow}$ is the abundance/peptide-matched null p that the set moves in its RRRM-2-predicted direction; “concord.” is the Spearman correlation between RRRM-2 RNA and OSD-462 protein effects (p_c its matched-null p). No targeted set is concordant in the predicted direction; ECM matrix proteins move significantly *opposite* to the transcript prediction (two-sided matched-null $p = 0.012$).

3.3 Layer 2 — but NCC *activation* is specifically suppressed (**PASS**)

The phosphoproteomics workbook passes the hard checkpoint: NCC (*Slc12a3*), SPAK (*Stk39*) and OSR1 (*Oxsr1*) regulatory sites are all quantified. The NCC N-terminal regulatory threonine cluster — the canonical SPAK/OSR1 target that activates the cotransporter — is strongly down in flight (mean $-0.83 \log_2$; 7 sites with $p < 0.05$, composite site 65;68 at -1.56), while the NCC C-terminal sites are unchanged — a within-protein specificity that rules out a trivial abundance artifact. The upstream SPAK activating sites fall in parallel (mean -0.41 ; S383 at -0.79 , $p = 2.8e - 04$). Genome-wide the phosphoproteome is centered (median $+0.007$), and these NCC/SPAK sites sit in its extreme down tail (bottom ~ 0.0 – 0.7 percentile). Because total NCC protein is unchanged, the phospho-occupancy change is at least as large (Table 5, Fig. 1C, Fig. 2B).

Gene	peptides	RRRM-2 RNA	OSD-462 protein
<i>Slc12a3</i>	70	-0.232	0.089
<i>Wnk1</i>	19	-0.274	-0.024
<i>Wnk4</i>	5	-0.105	0.087
<i>Stk39</i>	12	0.052	0.081
<i>Oxsr1</i>	23	-0.072	0.016
<i>Calb1</i>	95	0.148	0.007
<i>Nedd4l</i>	31	-0.108	0.068
<i>Cul3</i>	58	-0.094	-0.001
<i>Fn1</i>	119	0.300	-0.006
<i>Col1a1</i>	23	0.486	-0.106
<i>Col3a1</i>	3	0.558	-0.216
<i>Fbn1</i>	24	0.571	-0.258

Table 4: Detection and flight effects for representative DCT/transport and matrix machinery (gene-level, both plexes). Peptide counts are summed across the two TMT plexes.

Gene	site	FL/GC	effect	95% CI	p	q	occ.
<i>Oxsr1</i>	339	10/10	0.005	[-0.05, 0.06]	0.867	0.921	-0.011
<i>Slc12a3</i>	65;68	10/10	-1.563	[-2.01, -1.12]	9.3×10^{-7}	4.7×10^{-5}	-1.652
<i>Slc12a3</i>	89	10/10	-0.930	[-1.30, -0.56]	6.6×10^{-5}	8.5×10^{-4}	-1.019
<i>Slc12a3</i>	53	10/10	-0.851	[-1.26, -0.44]	3.8×10^{-4}	0.002	-0.940
<i>Slc12a3</i>	68	10/10	-0.794	[-1.17, -0.42]	3.3×10^{-4}	0.002	-0.883
<i>Slc12a3</i>	65	10/10	-0.790	[-1.04, -0.54]	3.5×10^{-6}	8.8×10^{-5}	-0.879
<i>Slc12a3</i>	58;65	10/10	-0.545	[-0.84, -0.25]	0.001	0.005	-0.634
<i>Slc12a3</i>	53;65	10/10	-0.350	[-0.50, -0.20]	9.7×10^{-5}	9.9×10^{-4}	-0.439
<i>Slc12a3</i>	53;68	5/5	-0.133	[-0.76, 0.49]	0.638	0.816	-0.222
<i>Slc12a3</i>	56;65	5/5	-0.124	[-0.56, 0.31]	0.530	0.769	-0.213
<i>Slc12a3</i>	120	10/10	0.027	[-0.17, 0.23]	0.780	0.884	-0.062
<i>Slc12a3</i>	120;124	10/10	0.039	[-0.16, 0.24]	0.688	0.816	-0.049
<i>Slc12a3</i>	122	10/10	0.045	[-0.17, 0.26]	0.665	0.816	-0.044
<i>Slc12a3</i>	96	10/10	0.048	[-0.09, 0.19]	0.480	0.741	-0.041
<i>Slc12a3</i>	124	10/10	0.051	[-0.25, 0.36]	0.727	0.843	-0.038
<i>Slc12a3</i>	122;124	10/10	0.168	[-0.06, 0.40]	0.141	0.257	0.080
<i>Slc12a3</i>	120;122	10/10	0.191	[-0.05, 0.43]	0.107	0.207	0.102
<i>Stk39</i>	382	5/5	-0.829	[-1.77, 0.11]	0.076	0.169	-0.910
<i>Stk39</i>	383	10/10	-0.793	[-1.16, -0.43]	2.8×10^{-4}	0.002	-0.874
<i>Stk39</i>	366	10/10	-0.520	[-0.71, -0.33]	1.7×10^{-5}	3.0×10^{-4}	-0.601
<i>Stk39</i>	382;383	10/10	-0.368	[-0.58, -0.16]	0.002	0.008	-0.449
<i>Stk39</i>	397	10/10	0.073	[-0.01, 0.16]	0.092	0.188	-0.008

Table 5: Layer 2 WNK-SPAK/OSR1-NCC phosphosite flight effects (FL–GC, $\log_2$; plex-adjusted per-site model). “occ.” is the protein-abundance-normalized phospho-occupancy effect (phospho – total protein). The NCC N-terminal regulatory threonine cluster (53–89) and SPAK activating sites (366, 382/383) are strongly down; NCC C-terminal sites (96–124) and OSR1-339 are flat.

3.4 Layer 3 — network candidates do not translate (**not supported**)

RRRM-2 LIONESS/node2vec top composite candidates are not enriched among OSD-462 protein- or phospho-changing genes at any cutoff (protein $p \geq 0.23$, phospho $p \geq 0.25$; Table 6, Fig. 1D). The network layer therefore remains a transcript-level candidate-prioritization device with no independent proteomic support.

candidates	n_p	obs	null	p	n_ϕ	obs	null	p
top-25	14	0.103	0.081	0.227	8	0.205	0.218	0.592
top-50	22	0.088	0.079	0.329	19	0.229	0.201	0.254
top-100	41	0.089	0.080	0.298	33	0.182	0.201	0.751

Table 6: Layer 3 network-candidate translation. For RRRM-2 LIONESS/node2vec top- k composite candidates: mean |protein effect| (n_p candidates) and mean per-gene max |phospho effect| (n_ϕ candidates) versus the abundance/detectability-matched null. Candidates are *not* enriched at either layer.

4 Cross-layer integration and decision

The pre-registered decision table resolves to: “*Protein-abundance concordance NULL for both matrix and DCT; DCT signal relocates to the phospho/activity layer.*” The honest reading across layers is two-pronged:

- **Matrix arm:** transcriptional only. The ECM program is induced at the RNA level in both cohorts but matrix protein abundance is not increased (it is mildly, significantly decreased), indicating transcript–protein uncoupling at terminal collection rather than net matrix deposition.
- **DCT/NCC arm:** an activity/signaling change. NCC transcript is down and total NCC protein is preserved, but the SPAK→NCC activating phosphorylation arm is specifically suppressed. This both *explains* why the DCT abundance test was null and *upgrades* the DCT claim to a signaling-level statement.

5 Biological implications

The WNK–SPAK/OSR1–NCC cascade is the master switch for Na–Cl reabsorption in the distal convoluted tubule; phosphorylation of the NCC N-terminal threonines by SPAK/OSR1 is the activating step, and SPAK is itself activated by WNK-dependent phosphorylation. Finding the NCC activating cluster and the SPAK activating sites coordinately among the most down-regulated phosphosites in the flight kidney — with total NCC protein unchanged — is direct molecular evidence that spaceflight reduces NCC *transport-pathway activation*, not transporter abundance. This is mechanistically more specific than the original transcriptomic observation and connects to the physiological context that motivated the axis (distal Na–Cl handling, K^+/Ca^{2+} balance, volume regulation, and renal stone risk). It also reframes the matrix result: the “matrix-high” signature is, in this cohort, a transcriptional program not yet realised as protein, consistent with active remodeling (transcription plus proteolysis) rather than fibrotic accumulation. Finally, the failure of network-nominated candidates to translate cautions that LIONESS/node2vec rewiring scores prioritise transcripts, not protein-level biology.

6 What this can and cannot establish

Can: that the RNA remodeling *direction* reproduces in a second, independent, cross-strain flight cohort at the RNA level; that the specific WNK–SPAK–NCC activation arm is sup-

OSD-462 / RR-10 multi-omics anchor — protein & phosphoprotein test of the RRRM-2 matrix-high / DCT-low remodeling signal

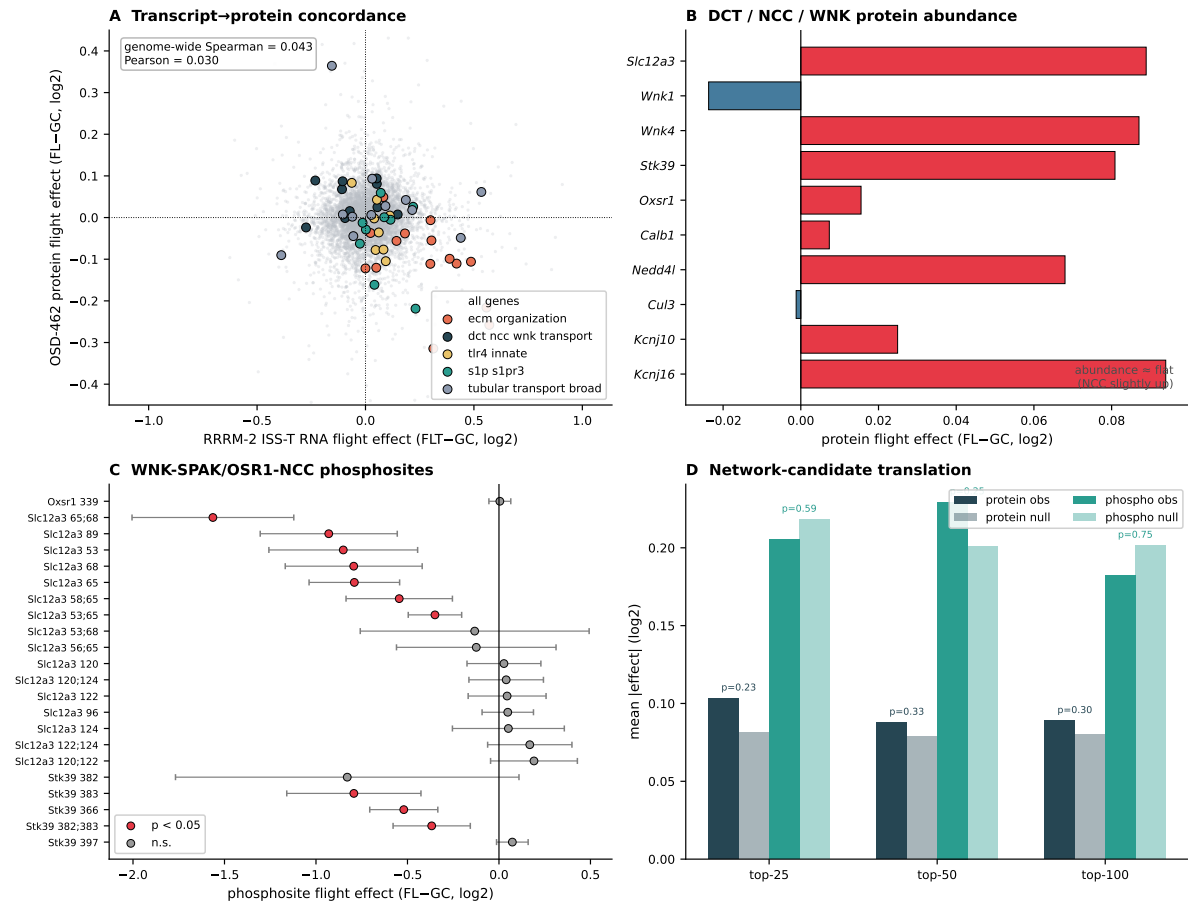


Figure 1: Multi-omics anchor dashboard. **A** Transcript (RRRM-2 ISS-T RNA) vs. OSD-462 protein flight effects; targeted sets coloured. Matrix genes (orange) occupy the transcript-up / protein-down quadrant. **B** DCT/NCC/WNK protein abundances are flat (NCC slightly up). **C** WNK–SPAK/OSR1–NCC phosphosite effects with 95% CIs; the NCC N-terminal cluster and SPAK activating sites (red, $p < 0.05$) are strongly down while C-terminal sites are flat. **D** Network-candidate translation: observed mean |effect| tracks the matched null at protein and phospho layers.

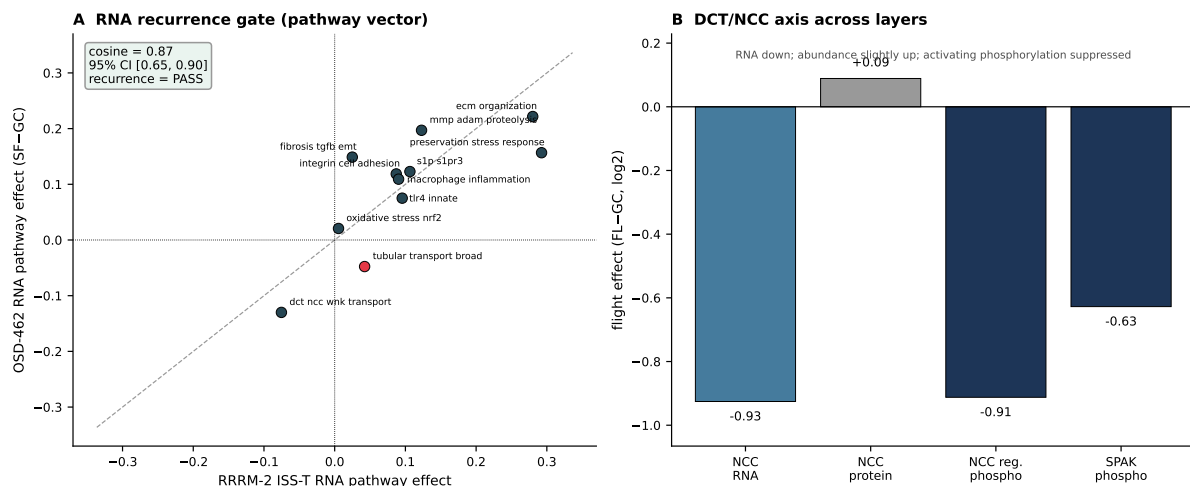


Figure 2: A RNA recurrence gate: OSD-462 vs. RRRM-2 pathway-effect vectors lie along the identity line (cosine 0.87, PASS); *dct_ncc_wnk_transport* sits in the shared-down quadrant, ECM/matrix in the shared-up quadrant. **B** The DCT/NCC axis across layers: NCC transcript down, total NCC protein maintained, NCC regulatory phosphorylation and SPAK phosphorylation strongly down — an activity-level, not abundance-level, change.

pressed at the phosphoprotein level; and that matrix induction is transcript-level only. **Can-**
not: physiology. There is no histology and no urine/serum electrolyte data in scope; even perfect concordance would yield a molecular claim (“reduced NCC transport-pathway activation”), not a functional one (“kidney Na-Cl handling changed”). A single terminal time point cannot resolve the transcript→protein timing that would explain the matrix uncoupling. The result does not rehabilitate “network rewiring” as a mechanism-discovery method. Histology and urine/serum chemistry remain explicit future work.

7 Reproducibility

All numbers and tables in this document are generated directly from the analysis outputs under `data/results/run_20260521_osd462_anchor/osd462_anchor/`: harmonized flight effects (`osd462_flight_effects.tsv`), RNA recurrence (`osd462_rna_recurrence.tsv`), protein concordance (`protein_concordance_summary.tsv`), phospho axis (`phospho_axis_summary.tsv`, `phospho_axis_verdict.tsv`), network translation (`network_translation.tsv`), and `results_summary.json`. Each layer writes a manifest with input SHA-256 digests; the run-level `manifest.json` references them. Pipeline: `scripts/osd462/00_harmonize.py` → `04_rna_recurrence.py` (gate) → `01_protein_concordance.py` → `02_phospho_axis.py` → `03_network_translation.py` → `05_plot_dashboard.py` → `06_compile_summary.py` → `07_build_compendium.py`; core estimators in `src/multiomics/osd462_anchor.py` with unit tests in `tests/test_osd462_anchor.py`. Matched nulls use 10000 draws (seed 20260521); the RNA recurrence bootstrap uses 2000 re-samples.